Week 7 Lab Assistant Preparation

IoT System with Raspberry Pi, MQTT, and HiveMQ

1. What Week 7 is about

Week 7 is about building a basic IoT system.

Students will use a Raspberry Pi to read the Raspberry Pi temperature, create a cloud MQTT broker using HiveMQ Cloud, publish messages from Python to the broker, subscribe to messages from Python, and combine everything into a small IoT system where the Raspberry Pi temperature is sent through MQTT.

> A device collects data, sends it to the internet, and another device or service can read that data.

In this lab, the Raspberry Pi is the IoT device, the temperature is the sensor data, HiveMQ Cloud is the message broker, and Python is used to send and receive messages.

2. Relevant repo folders/files

The relevant folder is:

```
week07/
```

Files inside:

```
week07/README.md
week07/01_temperature.md
week07/02_cloud_service.md
week07/03_python_pub_sub.md
week07/04_iot_service.md
```

README.md

Very short overview: Create your own IoT system. It only says students will program their own IoT system.

01_temperature.md

Students learn how to read Raspberry Pi temperature:

```
vcgencmd measure_temp
```

Then they create a Python script using subprocess to run that command.

02_cloud_service.md

Students create a HiveMQ Cloud MQTT service:

- Create account
- Create cluster
- Create MQTT username/password
- Get host and port
- Test with HiveMQ Web Client

03_python_pub_sub.md

Students install paho-mqtt and create a Python publisher and subscriber. The publisher sends messages to HiveMQ. The subscriber listens for messages.

04_iot_service.md

Students combine everything: read Raspberry Pi temperature, publish/read it through MQTT, submit Python files and reflection.

3. What students are expected to do

- Open terminal on the Raspberry Pi.
- Run vcgencmd measure_temp.
- Create a Python file that reads the Pi temperature.
- Create a HiveMQ Cloud account.
- Create an MQTT cluster.
- Create MQTT credentials.
- Use host, port, username, and password in Python.
- Install the MQTT Python library.
- Run a publisher script.
- Run a subscriber script.
- Use two terminals: one for subscriber and one for publisher.
- Send Raspberry Pi temperature data through MQTT.
- Take screenshots for evidence.
- Write a short reflection.

```
pip3 install paho-mqtt
```

4. What you should personally understand before the lab

A. What IoT means

IoT means Internet of Things.

> IoT is when a physical device, like a Raspberry Pi, sensor, camera, or smart light, connects to the internet and sends or receives data.

Examples include smart thermostats, smart watches, security cameras, and Raspberry Pi monitoring.

```
Raspberry Pi temperature -> Python -> MQTT broker -> subscriber/web client
```

B. What MQTT is

MQTT is a lightweight messaging protocol often used in IoT.

> MQTT is like a message delivery system. Devices do not send messages directly to each other. They send messages to a broker, and the broker passes them to anyone listening.

Term	Meaning
Publisher	Sends a message
Subscriber	Receives messages
Broker	Middle server that handles messages
Topic	Message channel/name
Payload	Actual message/data

```
Topic: pi/temperature
Payload: 48.2
```

Publisher sends temperature to pi/temperature. Subscriber listens to pi/temperature.

C. What HiveMQ Cloud is

> HiveMQ Cloud is the internet server that receives MQTT messages and sends them to subscribers.

Students need host, port, username, and password. For HiveMQ Cloud, the secure MQTT port is usually 8883.

D. What TLS means here

TLS means secure encrypted connection. Students do not need deep theory.

> TLS is what makes the connection secure, similar to HTTPS for websites.

```
ssl.SSLContext(...)
port = 8883
```

E. What vcgencmd measure_temp does

This command reads the Raspberry Pi CPU temperature. Example output:

```
temp=47.8'C
```

Python can run the terminal command from inside Python:

```
subprocess.check_output(["vcgencmd", "measure_temp"])
```

Important: this command works on Raspberry Pi OS. It may not work on a normal laptop.

5. Common student mistakes and how to fix them

Mistake 1: Running vcgencmd on a non-Raspberry Pi

Problem: vcgencmd: command not found. Reason: they are not on the Raspberry Pi, or Raspberry Pi tools are missing.

```
ssh pi_username@pi_ip_address
vcgencmd measure_temp
```

Mistake 2: Typo in vcgencmd

Students may type the wrong command. The correct command is:

```
vcgencmd measure_temp
```

Mistake 3: Python file has wrong indentation

Students may remove indentation after def. Everything inside the function must be indented.

```
def read_temp():
    output = subprocess.check_output(["vcgencmd", "measure_temp"])
    temp_str = output.decode()
    temp_value = float(temp_str.replace("temp=", "").replace("'C", ""))
    return temp_value
```

Mistake 4: They forget to install paho-mqtt

Error: ModuleNotFoundError: No module named paho. Fix by installing the package.

```
python3 -m pip install paho-mqtt
python3 my_publisher.py
```

Mistake 5: Wrong HiveMQ credentials

Check host, port 8883, username, password, and TLS. Tell students not to include spaces when copying credentials.

Mistake 6: Using the wrong hostname

Wrong: https://console.hivemq.cloud. Correct: the cluster host, usually something like yourclusterid.s1.eu.hivemq.cloud.

Mistake 7: Subscriber is not running before publisher

Run the subscriber first and leave it open. Then run the publisher in a second terminal.

Mistake 8: Topic mismatch

MQTT topics must match exactly and are case-sensitive.

```
pi/temperature
```

Mistake 9: Confusion between topic and payload

Topic is where the message goes. Payload is the actual message.

```
topic = "pi/temperature"
payload = "48.2"
```

Mistake 10: Password exposed in screenshots

Tell students not to show passwords in public screenshots or GitHub repos.

6. Commands, tools, and concepts you should be ready to explain

Terminal commands

Check Raspberry Pi temperature

```
vcgencmd measure_temp
```

Create Python file

```
nano read_temp.py
```

Run Python file

```
python3 read_temp.py
```

Install MQTT library

```
python3 -m pip install paho-mqtt
```

Check Python version

```
python3 --version
```

Check network connection

```
ping google.com
```

Check Pi IP address

```
hostname -I
```

Python and MQTT concepts

subprocess

Used to run terminal commands from Python.

.decode()

Converts command output from bytes into normal text.

.replace()

Removes unwanted text so the temperature can be converted to a float.

MQTT publisher

The sender.

MQTT subscriber

The listener.

MQTT broker

The middle server, like a post office.

7. Physical or software setup to check before the session

Hardware checklist

- Raspberry Pis available
- Power supplies available
- MicroSD cards inserted
- Pi can boot
- Keyboard/mouse/monitor available if needed
- Wi-Fi or Ethernet available
- Students can access terminal

If using SSH

- Pi and student laptops are on same network
- SSH is enabled
- Students know username/password
- Students know Pi IP address

Internet checklist

This lab needs internet because HiveMQ Cloud is online. Check:

```
ping google.com
ping hivemq.com
```

Python setup checklist

```
python3 --version
pip3 --version
python3 -m pip install paho-mqtt
python3 -c "import paho.mqtt.client; print('paho works')"
```

HiveMQ setup checklist

- Go to HiveMQ Cloud console.
- Sign up or log in.
- Create cluster.
- Create credentials.
- Copy host.
- Use port 8883.
- Open Web Client.
- Subscribe to a topic.

- Publish a test message.

Potential problem: account creation may require email verification, so students may get delayed if they cannot access email.

8. Simple explanations you can give to beginners

Explain IoT

> IoT means connecting real-world devices to the internet. In this lab, the Raspberry Pi is the device, and its temperature is the data.

Explain MQTT

> MQTT is a simple way for devices to send messages. One device publishes a message, another subscribes to receive it, and a broker sits in the middle.

Explain Topic

> A topic is like a channel name. If I publish to pi/temperature, anyone subscribed to pi/temperature can receive the message.

Explain Payload

> Payload is the actual content of the message. If the topic is pi/temperature, the payload might be 48.2.

Explain Broker

> The broker is like a middleman. Devices do not need to know each other directly. They only need to connect to the broker.

Explain Publisher/subscriber

> Publisher sends data. Subscriber waits and receives data.

Explain Why two terminals are needed

> One terminal keeps listening. The other terminal sends the message. If the listener is not open, you may miss the message.

Explain Raspberry Pi temperature

> The Raspberry Pi has a command that tells us how hot the CPU is. We use Python to read that value and send it to the cloud.

9. Suggested final IoT Python structure

The repo asks students to combine temperature reading with MQTT. A simple final structure could be:

```
read_temp.py
publisher.py
subscriber.py
```

Or one file:

```
temperature_publisher.py
```

The final publisher should read temperature, convert it to text, and publish it to an MQTT topic.

```
Topic: pi/temperature
Payload: 48.2
Alternative payload: 48.2 C
```

10. Likely flow of the lab session

Stage 1 - Read temperature

```
vcgencmd measure_temp
# Expected output:
temp=47.8'C

python3 read_temp.py
# Expected output:
Temperature: 47.8
```

Stage 2 - Create HiveMQ Cloud broker

- HiveMQ Cloud account
- MQTT cluster
- MQTT credentials
- HOSTNAME, USERNAME, PASSWORD, PORT = 8883

Stage 3 - Test using Web Client

```
Subscribe to: test/topic
Publish: hello
```

Stage 4 - Python publisher

```
python3 my_publisher.py
```

Stage 5 - Python subscriber

```
Terminal 1:
python3 my_subscriber.py

Terminal 2:
python3 my_publisher.py
```

Stage 6 - Final IoT system

```
Raspberry Pi reads temperature
Python publishes temperature to HiveMQ
Subscriber or Web Client receives temperature
```

11. Short checklist for you before the lab

- [ ] Raspberry Pis boot properly
- [ ] Internet is working
- [ ] Students can access terminal
- [ ] SSH works if needed
- [ ] vcgencmd measure_temp works
- [ ] Python 3 is installed
- [ ] pip3 works
- [ ] paho-mqtt can be installed
- [ ] HiveMQ Cloud website is accessible
- [ ] You know how to create a cluster
- [ ] You know where to find host, port, username, password
- [ ] You can explain publisher, subscriber, broker, topic, payload
- [ ] You can help students run two terminals

- [ ] You remind students not to expose passwords in screenshots/GitHub

12. Missing or incomplete parts in the repo

1. 01_temperature.md has an incomplete sentence

It says: Submit an image with the. This is unfinished. Most likely students should submit a screenshot of terminal output from vcgencmd measure_temp, but staff may need to clarify.

2. Spelling mistakes in repo instructions

Examples include temomether, screeshot, Rasberry Pi, and terminal.s. Not serious technically, but students may notice.

3. No full final solution template

04_iot_service.md tells students to combine everything, but it does not provide a clear starter template. Be ready to guide them on combining read_temp() with MQTT publishing.

4. Credentials security is not discussed

The repo tells students to put username/password directly into code. For a beginner lab this is acceptable, but warn them not to push real passwords to public GitHub repositories.

5. HiveMQ UI may have changed

The exact buttons may differ from the repo. Be ready to help students find equivalent options for Access Management, Credentials, Web Client, and Create Cluster.

6. Cloud setup depends on email/account access

Students may get stuck because they need email verification, Google login, or account creation.

13. What you should be ready to demonstrate live

```
vcgencmd measure_temp
python3 read_temp.py

# Terminal 1
python3 my_subscriber.py

# Terminal 2
python3 my_publisher.py
```

> The publisher sent a message to HiveMQ. The subscriber received it because it was listening to the same broker/topic.

14. Main beginner confusion points

- HiveMQ console vs MQTT host
- Username/password for website vs MQTT credentials
- Topic vs message
- Publisher vs subscriber
- Broker vs client
- Terminal command vs Python code
- Running code on laptop vs running code on Raspberry Pi

> The HiveMQ website login is not necessarily the same as the MQTT username/password. They must create MQTT credentials inside the cluster.

15. Very simple final explanation of the whole lab

Today we are making a very small IoT system. The Raspberry Pi will measure its own temperature. Then Python will send that temperature to a cloud MQTT broker called HiveMQ. Another Python program or the HiveMQ Web Client will listen for that message and display it. This is the same basic idea used in real IoT systems like smart homes, sensors, and monitoring devices.